The importance of plants in ecosystems

Plants are a very important part of ecosystems for a number of reasons. One reason is that they are producers, which means they make their own food from inorganic (non-living) ingredients, and other living things consume this food. Another important reason is that as they are making food they take in **carbon dioxide** and release **oxygen**.

Green plants take in carbon dioxide gas through their leaves. Carbon dioxide, water and sunlight are used during a process called **photosynthesis** to help the plant produce a sugary food called glucose. When the plant breaks down water during photosynthesis, one of the waste products is oxygen. The plant releases the oxygen through its leaves. Most animals, including humans, need oxygen in the air they breathe for **respiration**.

The word equation for photosynthesis is:

carbon dioxide + water (+ light energy) = glucose + oxygen

The chemical formula for carbon dioxide gas is CO_2 (one part carbon and two parts oxygen).

The chemical formula for water is H_2O (two parts hydrogen and one part oxygen).

☞ **1** Write the answers to these questions in the spaces provided.

a) What is the name of the gas that plants take in? _____

b) Which parts of the plant collect the gas? _____

c) In what process is the gas used by the plant? _____

d) What two substances result from this process? _____

e) One of these substances is a waste product, not used by the plant. Which substance is this? _____

f) Why do most animals need oxygen? _____

g) Which chemical element is represented by C? _____

h) Which chemical compound is CO_2? _____

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The importance of plants in ecosystems CONT'D

i) What common liquid is H_2O ? _____

j) Which two elements make up water molecules? _____

2 What would be some of the effects on a forest ecosystem if more than half the trees and plants were cut down or destroyed? List five effects, and underline who or what would be affected. Share your answers with a partner. One has been done for you.

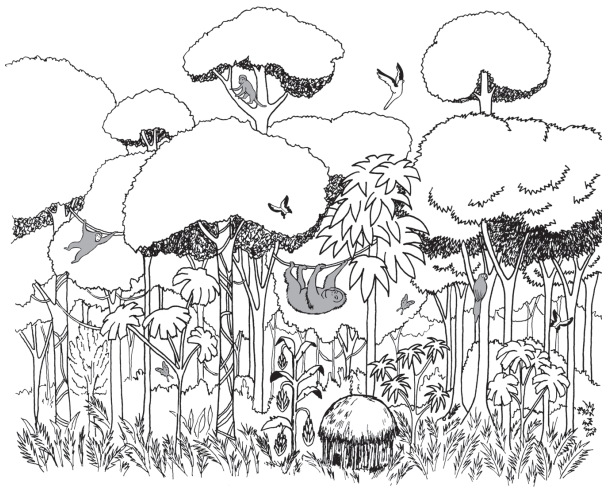
Birds such as woodpeckers would lose their nests or homes.

- _____
- _____
- _____
- _____
- _____

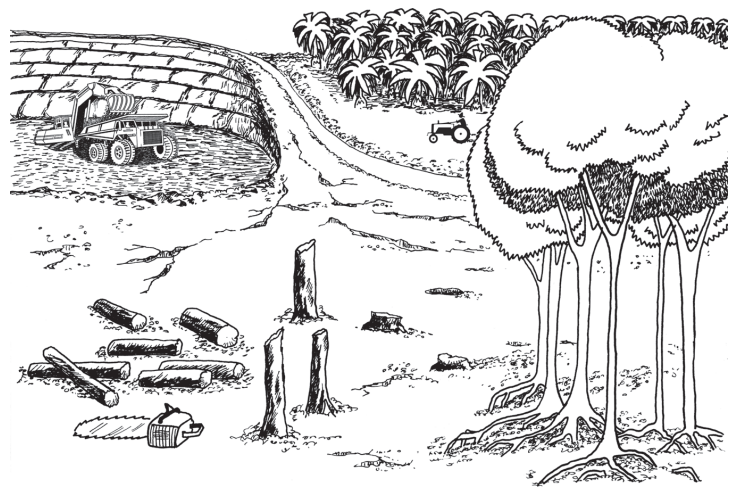
3 According to this Activity sheet, what are the two main reasons why plants are important for ecosystems?

_____

_____



Before logging



After logging

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